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N-gram Activity Sheet – Answers
DSA03: Natural Language Processing

Corpus (19 tokens): the cat sat on the mat . the dog sat on the rug . the cat ate fish .

Counts used (from the corpus)

Unigram counts (N = 19): the = 5, cat = 2, sat = 2, on = 2, "." = 3, mat = 1, dog = 1, rug = 1, ate = 1, fish = 1.

Bigram counts: the cat = 2, cat sat = 1, sat on = 2, on the = 2, the mat = 1, mat . = 1, . the = 2, the dog = 1, dog sat = 1, the rug = 1, rug . = 1, cat ate = 1, ate fish = 1, fish . = 1.

Trigram counts: the cat sat = 1, cat sat on = 1, sat on the = 2, on the mat = 1, the mat . = 1, mat . the = 1, . the dog = 1, the dog sat = 1, dog sat on = 1, on the rug = 1, the rug . = 1, rug . the = 1, . the cat = 1, the cat ate = 1, cat ate fish = 1, ate fish . = 1.

Activity 1: Backoff (penalty 0.4)

Method: if the trigram was seen, score = count(w1 w2 w3) / count(w1 w2). Otherwise back off to the bigram: score = 0.4 × count(w2 w3) / count(w2). If the bigram is also unseen, back off to the unigram: score = 0.4 × 0.4 × count(w3) / N.

a) "the cat sat" – trigram level

The trigram "the cat sat" occurs 1 time and "the cat" occurs 2 times.

Score = 1/2 = 0.5 (level: trigram)

b) "dog ate fish" – bigram level

The trigram "dog ate fish" has count 0, so back off. The bigram "ate fish" occurs 1 time and "ate" occurs 1 time, so P(fish | ate) = 1/1 = 1.

Score = 0.4 × 1 = 0.4 (level: bigram)

c) "dog sat fish" – unigram level

The trigram "dog sat fish" has count 0 and the bigram "sat fish" has count 0, so back off to the unigram. P(fish) = 1/19.

Score = 0.4 × 0.4 × 1/19 = 0.16/19 ≈ 0.0084 (level: unigram)

Activity 2: Deleted interpolation

Formula: $P(w3 \mid w1\ w2) = \lambda1 \cdot P(w3) + \lambda2 \cdot P(w3 \mid w2) + \lambda3 \cdot P(w3 \mid w1\ w2)$, where $\lambda1 + \lambda2 + \lambda3 = 1$.

Part A: P(sat | the cat) with $\lambda1 = 0.1$, $\lambda2 = 0.3$, $\lambda3 = 0.6$

P(sat) = 2/19 ≈ 0.1053

P(sat | cat) = count(cat sat)/count(cat) = 1/2 = 0.5

P(sat | the cat) = count(the cat sat)/count(the cat) = 1/2 = 0.5

P = 0.1(0.1053) + 0.3(0.5) + 0.6(0.5) = 0.0105 + 0.15 + 0.30

P(sat | the cat) ≈ 0.4605

Part B: held-out sentence "the dog ate fish"

Trigrams: "the dog ate" and "dog ate fish". Needed values:

• P(ate) = 1/19 ≈ 0.0526; P(ate | dog) = 0/1 = 0; P(ate | the dog) = 0/1 = 0

• P(fish) = 1/19 ≈ 0.0526; P(fish | ate) = 1/1 = 1; P(fish | dog ate) = 0 ("dog ate" never appears, so the trigram term contributes 0)

	P(ate | the dog)	**P(fish | dog ate)**	**Sum of ln**
Set 1 (0.1, 0.3, 0.6)	0.1(0.0526) = 0.00526	0.1(0.0526)+0.3(1) = 0.30526	ln(0.00526) + ln(0.30526) = −5.2470 + (−1.1866) = −6.434
Set 2 (0.3, 0.6, 0.1)	0.3(0.0526) = 0.01579	0.3(0.0526)+0.6(1) = 0.61579	ln(0.01579) + ln(0.61579) = −4.1484 + (−0.4849) = −4.633

Choice: Set 2.

Deleted interpolation picks the λ set that maximises the (log) likelihood of the held-out data. Set 2 gives −4.633, which is higher (less negative) than Set 1's −6.434. Both held-out trigrams are unseen as trigrams, so the trigram term contributes nothing, and the bigram term is what carries the probability (e.g. P(fish | ate) = 1). Set 2 gives the bigram the largest weight (0.6) and the trigram the smallest (0.1), so it fits the held-out sentence better.

Activity 3: Entropy and perplexity

Formulas: $H = -\Sigma\, p \log_2 p$ and Perplexity = 2^H.

Part A: Sunny 0.5, Cloudy 0.25, Rainy 0.25

$H = -[0.5 \log_2 0.5 + 0.25 \log_2 0.25 + 0.25 \log_2 0.25]$

$H = -[0.5(-1) + 0.25(-2) + 0.25(-2)] = 0.5 + 0.5 + 0.5$

Entropy H = 1.5 bits

Perplexity = 2^1.5 ≈ 2.83

Part B: next word after "the"

"the" appears 5 times, followed by: cat (2 times), mat (1), dog (1), rug (1).

Next word	**Count**	**Probability**
cat	2	2/5 = 0.4
mat	1	1/5 = 0.2
dog	1	1/5 = 0.2
rug	1	1/5 = 0.2

$H = -[0.4 \log_2 0.4 + 3 \times (0.2 \log_2 0.2)]$

$H = -[0.4(-1.3219) + 3 \times 0.2(-2.3219)] = 0.5288 + 1.3932$

Entropy H ≈ 1.92 bits

Perplexity = 2^1.92 ≈ 3.79

Entropy of the next word after "sat"

"sat" appears 2 times and is followed by "on" both times, so P(on | sat) = 1.

$H = -[1 \times \log_2 1] = 0$ bits (perplexity = 2^0 = 1; the next word is completely predictable)